

Dr. Benedikt Zönnchen

Curriculum Vitae

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Academic Career

- 10/2023 - today
2021
03/2016 - 07/2020
10/2013 - 02/2016
10/2010 - 09/2013
- Head of Research**, *Munich Center for Digital Sciences and AI*
Dr. rer. nat. in Informatics, summa cum laude, *Technical University of Munich*
Thesis: Efficient parallel algorithm for pedestrian simulation.
Supervised by Prof. Hans-Joachim Bungartz (TUM) & Prof. Gerta Köster (HM)
PhD Candidate, *Hochschule München University of Applied Sciences*
Informatics (M. Sc.), *Technical University of Munich*
Informatics (B. Sc.), *Hochschule München University of Applied Sciences*

Non-Academic Work Experience

- 09/2022 - 02/2023
02/2022 - 09/2023
08/2020 - 01/2022
03/2011 - 10/2011
09/2008 - 07/2009
09/2005 - 07/2008
- Lecturer**, *Hochschule München University of Applied Sciences*
Educational-Tech Engineer, *Hochschule München University of Applied Sciences*
Senior Advisor for Computer Science Education, *Hochschule München University of Applied Sciences*
Software Developer (student employee), *Prevero AG*
Web Developer, *Nokia Siemens Networks GmbH & Co. KG*
Software Developer (professional training), *Siemens AG*

Research Interests

- AI & Systems Theory** I explore human–AI co-creativity and co-evolution as emergent socio-technical processes, in which AI systems are shaped by feedback loops, develop operational closure, and draw on the contingencies generated through communication.
- AI in Culture, Arts and Education** I apply this lens to the fields of culture, arts, and education to examine how generative AI transforms creative practices, roles, and learning environments.

Other Interests

- Open Education** Free and open education, schooling and education in the digital era
Complex Systems How do complex system interact? What is the relation of entropy and complexity?
Philosophy Constructivism, Philosophy of Mind, Phenomenology, Existentialism, Philosophy of Language
Creative Coding Live Programming, Algorithmic Composition, Sound Design, Generative Design
Formal methods Automata Theory, Logic, Laws of Form

Recent Publications

- 2026 Dzhimova, M. & **Zönnchen, B.** (2026). From an experimental technology to a mainstream product: Implications for generative AI as an artistic medium. In *Artificial Intelligence in Creative Industries: Psychological and Social Implications*. 10.4324/9781003606802-8
- 2025 **Zönnchen, B.**, Dzhimova, M. & Socher, G. (2025). From intelligence to autopoiesis: rethinking artificial intelligence through systems theory. *Frontiers in Communication*. Vol. 10. 10.3389/fcomm.2025.1585321
- 2025 **Zönnchen, B.**, Hobelsberger, M., Socher, G., Thurner, V. & Ottinger, S. (2025). Exploring the role of large language models as artificial tutors. In *Proceedings of the 2025 IEEE Education Engineering Conference*. 10.1109/EDUCON62633.2025.11016571

Recent Talks

- 2025 **Introduction to Artificial Intelligence and Machine Learning**, Artificial Intelligence in Culture and Arts (AICA), Wavelab, [link](#)
- 2024 **A Systemic View on Society**, Munich Creative Business Week 2024, [link](#)
- 2024 **AI as a Tool of Emancipation?**, IGA Convention
- 2023 **Generative AI in the Context of XR**, XR for the middle class, IHK Munich, [link](#)

2023 **Generative AI: Between Tool and Communication Partner**, Ready for the future: AI-skills as part of the curriculum, Dialogue forum, HM

Practical Experience

Software Development & Machine Learning

Python	Expert , Python developer since 2016 (NumPy, Pandas, Django, SciPy), scripting, trainer for Python, author of an open textbook (Link)
Java	Expert , Java developer since 2006, main contributor to the open-source simulation framework Vadere (Link), language of my education at Siemens and the bachelor program
PyTorch, Scikit-learn	Expert , teaching, development of a melody generator (LSTM, Transformer)
spaCy, BERTopic	Proficient , rule-based and transformer-based topic text analysis and topic modeling
OpenCL	Proficient , GPU programming during my PhD project
JS, HTML, CSS, PHP	Proficient , contributor to a social media platform (1 year), p5.js visualizations (Link)
C/C++	Proficient , high performance computing during my PhD project
Database, SQL	Proficient , constant use during my work as software developer
Rust, Scala, Haskell	Basic , personal interest, music making with TidalCycles

Other Technologies

Git	Expert , Git user since 2011
L ^A T _E X	Expert , L ^A T _E X advocate since 2011
SuperCollider	Expert , DSP, sound design, live programming, author of an open textbook (Link)
Jupyter ecosystem	Proficient , deployment of a JupyterHub, Development of Jupyter notebooks for my students, author of an interactive Jupyter book
Docker, Kubernetes	Basic , usage in the context of education and scientific work

Teaching

Winter 2025/26	Sustainable AI , lecture (bachelor), <i>trainer and coordinator</i> , HM
Winter 2023/24	Artificial Intelligence in Culture and Arts , project-based workshop, <i>trainer</i> , HM & HMTM
Winter 2023/24	Sustainable AI , lecture (bachelor), <i>trainer and coordinator</i> , HM
Summer 2023	Artificial Intelligence in Culture and Arts , workshop, <i>Trainer</i> , HM & HMTM
Winter 2022/23	Sustainable AI , lecture (bachelor), <i>lecturer</i> , HM
Winter 2022/23	Computational Thinking , lecture (bachelor), <i>trainer</i> , HM
Winter 2021/22	Computational Thinking , lecture (bachelor), <i>trainer</i> , HM
Annually since 2020	Preparation for Computer Science , 5-day course (bachelor), <i>trainer and coordinator</i> , HM
Winter 2019/20	Machine Learning in Crowd Modeling and Simulation , <i>guest lecture</i> , TUM
Winter 2016/17	Linear Algebra , lecture (bachelor), <i>lecturer</i> , HM
Summer 2016	Scientific Computing , Seminar (bachelor), <i>lecturer</i> , HM
Summer 2016	Theoretical Computer Science , lecture (bachelor), <i>trainer</i> , HM

Scholarships & awards

2023	MINT Challenge award (Award for the course <i>Sustainable AI</i>)
2022	Dissertation award (Oskar-von-Miller Award)
2021	Dissertation award (Bund der Freunde der Technischen Universität München e. V.)
2012 – present	Alumni of the German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes) and the Max Weber-Program of the State of Bavaria (Max Weber-Programm Bayern)
2013	RiMEA sponsorship award, Valedictorian

Publication List

- 2026 Dzhimova, M. & **Zönnchen, B.** (2026). From an experimental technology to a mainstream product: Implications for generative AI as an artistic medium. In *Artificial Intelligence in Creative Industries: Psychological and Social Implications*. 10.4324/9781003606802-8
- 2025 **Zönnchen, B.**, Dzhimova, M. & Socher, G. (2025). From intelligence to autopoiesis: rethinking artificial intelligence through systems theory. *Frontiers in Communication*. Vol. 10. 10.3389/fcomm.2025.1585321
- 2025 **Zönnchen, B.**, Hobelsberger, M., Socher, G., Thurner, V. & Ottinger, S. (2025). Exploring the role of large language models as artificial tutors. In *Proceedings of the 2025 IEEE Education Engineering Conference*. 10.1109/EDUCON62633.2025.11016571
- 2025 Böttcher, A., Thurner, V. & **Zönnchen, B.** (2025). Concepts for teaching software development in the age of AI-tools. In *Proceedings of the 2025 IEEE Education Engineering Conference*. 10.1109/EDUCON62633.2025.11016505
- 2024 **Zönnchen, B.**, Thurner, V. & Böttcher, A. (2024). On the impact of ChatGPT on teaching and studying software engineering. In *Proceedings of the 2024 IEEE Education Engineering Conference*. 10.1109/EDUCON60312.2024.10578680
- 2024 **Zönnchen, B.**, Böhm, C. & Socher, G. (2024). Bridging disciplines in higher education: The convergence of AI and sustainability. In *Proceedings of the 10th International Conference on Higher Education Advances (HEAD'24)*. 10.4995/HEAD24.2024.17278
- 2024 Sanchez, T., **Zönnchen, B.**, Held, H., Dzhimova, M. & Socher, G. (2024). Evaluating interdisciplinary and project-based learning on AI in culture and arts. In *EDULEARN24 Proceedings*, 10.21125/edulearn.2024.0757
- 2024 **Zönnchen, B.**, Böhm, C., von Schwichow, H., Socher, G. & Wurster, S. (2024). Empowering interdisciplinary expertise: Evaluating the impact of challenge-based learning on AI and sustainability literacy in higher education. In *EDULEARN24 Proceedings*, 10.21125/edulearn.2024.1433
- 2024 Hammer, S., Ottinger, S., **Zönnchen, B.**, Hohendanner, M., Hobelsberger, M. & Thurner, V. (2024). ChatGPT in higher education: Perceptions of computer science-related students. In *Proceedings of the 2024 IEEE Education Engineering Conference*. <https://doi.org/10.1109/EDUCON60312.2024.10578742>
- 2023 **Zönnchen, B.**, Friedrich, M. & Thurner, V., (2023). Nachhaltigkeit in der informatischen Lehre am Beispiel KI. In *Tagungsband zum 5. Symposium zur Hochschullehre in den MINT-Fächern*, 10.57825/repo_in-4411
- 2022 Hammer, S., Ottinger, S., Thurner V. & **Zönnchen, B.**, (2022). Bonding in times of pandemia – a concept for purely virtual kick-off days to the student entry phase. In *Mobility for Smart Cities and Regional Development – Challenges for Higher Education*, 10.1007/978-3-030-93904-5_19
- 2020 **Zönnchen, B.**, & Köster, G., (2020). GPGPU computing for microscopic pedestrian simulation. In *Parallel Computing: Technology Trends*, 10.3233/APC200029
- 2020 **Zönnchen, B.**, Kleinmeier, B. & Köster, G., (2020). Vadere – a simulation framework to compare locomotion models. In *Traffic and Granular Flow 2019*, 10.1007/978-3-030-55973-1_41
- 2019 **Zönnchen, B.**, Kleinmeier, B., Gödel, M., & Köster, G., (2019). Vadere: an open-source simulation framework to promote interdisciplinary understanding. In *Collective Dynamics*, 4, 10.17815/CD.2019.21
- 2019 **Zönnchen, B.**, Laubinger, M., & Köster, G., (2019) Towards faster navigation algorithms on floor fields. In *Traffic and Granular Flow '17*, 10.1007/978-3-030-11440-4_34
- 2018 **Zönnchen, B.**, & Köster, G., (2018) A parallel generator for sparse unstructured meshes to solve the eikonal equation. In *Journal of Computational Science*, 10.1016/j.jocs.2018.09.009
- 2016 **Zönnchen, B.**, & Gerta Köster, (2016). Detecting arbitrarily shaped queues using the fast marching method. *8th International Conference on Pedestrian and Evacuation Dynamics*, Hefei, China
- 2015 Köster, G. & **Zönnchen, B.**, (2015). A queuing model based on social attitudes. In *Traffic and Granular Flow '15*, 10.1007/978-3-319-33482-0
- 2014 Köster, G., & **Zönnchen, B.**, (2014). Queuing at bottlenecks using a dynamic floor field for navigation. In *Transportation Research Procedia*, 10.1016/j.trpro.2014.09.029